problem, Davy, in England, demonstrated that two pieces of ice rubbed together in a vacuum would produce heat enough to melt the ice. Obviously the heat could not have come from the ice, but it could not come from the air, as that possible source had been eliminated by the vacuum. Where did it come from? Rumford thought it was produced by friction, and, in order to demonstrate the truth of his theory, secured a large piece of brass, hollowed it at one end, and fitted into it a cylinder of steel weighing 5 tons. By an ingenious mechanical contrivance attached to the cylinder, it was made to revolve within the brass tube thirty-two times a minute. Water was placed in a box around the brass, and at the end of two hours and a half the water actually boiled. Water had been transformed into heat without the shadow of a doubt. This experiment was, of course, a death blow to the theory that heat was a fluid, and was the absolute foundation of the belief that heat is a form of motion.

In 1843. Dr. Joule, of Manchester, England, carried out his classic experiment which revealed to the world the mechanical equivalent of heat. He took a paddle and made it revolve with as little friction as possible in a vessel containing a pound of water whose temperature was known. The paddle was actuated by a known weight falling through a known distance. A pound falling through a distance of 1 foot gives a foot pound of energy. At the beginning of the experiment a thermometer was placed in the water and the temperature noted. The paddle was made to revolve by the falling weight, and when 722 foot-pounds of energy had been expended on the pound of water the temperature had risen 1 degree and the relationship between heat and mechanical work was found. Joule used mercury instead of water later, and verified his original discovery. This simply means that the amount of energy required to raise a pound of water 1 degree of Fahrenheit is sufficient to raise it 772 feet against the force of gravity. Later more careful experiments determined that the correct figure was 778 pounds. Therefore, 778 foot-pounds represents one British thermal unit.

BEST METHOD OF CARING FOR FLUES WHILE OUT OF BOILERS.

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This is an old subject, and has been threshed out a great many times. Still, I find there are a number of roads that do not appear to make use of the best practice in caring for flues and preparing same ready to go back in the boiler.

I notice there are but a few roads that weld on what we call a safe-end piece firebox end. When I say we I refer to the Chicago, Milwaukee & St. Paul Railway Company. The majority of all the flues we buy have a swede iron safe end welded on each flue. This swede iron safe end is more like a good steel tube, and is much better quality than the balance of the flue or what we term body grade, and will stand much more service and abuse at the firebox end than the ordinary body grade flue which a great number of roads are using, applying no safe end whatever, and when welding up flues they cut up this same body grade for piecing, which should not be termed safe ending.

And just think! Some shops are cutting these same pieces off with an old-style wheel cutter or square-nose tool, and then they make no attempt to remove the burr from the firebox end whatever. This prevents the flue end passing over the mandrel on the flue welder as readily as it should, and due to this fact the mandrel is some smaller than it should be; of course, this allows the flue to be contracted some at each weld or below the original size. Again, if the burr was taken off, as it should be, a larger mandrel could be used on the flue welder, and the flue would turn over for beading much easier and better with less liability of cracking and less chance for feather edge or burr around the bead after calking.

For welding 2-inch flues we are using a 1 11/16-inch mandrel; this is about the original inside diameter of the flue. We also heat and open up all flues at the front end to fit the hole in the front flue sheet; this does away with the shimming, which takes time and is poor practice.

Every weld on a flue is one more chance for a failure, because many times flues are slightly overheated at the weld and become quite thin just back of the weld. Leaks do develop at these points, and flues do break in two at the weld many times when being rattled or at other times in service, so I say the less welds you have on a flue the better. After a flue is rattled or cleaned it is our practice to inspect the flue thoroughly, and if there is any indication of the flue being light the flues are weighed up. A 2-inch flue weighing less than 1 2/3 pounds to the foot is scrapped, and a 2¼-inch flue must weigh 2 pounds to the foot, and a flue that has five or six or more welds on might weigh up all right, but the old part of the flue would not be good for further service.

Then, again, it is the practice of a great many roads to apply these pieces or weld on but one end of a flue. This I do not consider good practice. Many times you have a flue with five, six and seven welds, and many times when you watch the flue rattler when flues are taken out after being rattled you will find that flues are badly split up or jammed up on the front end, due to the quality and lightness of the flue. This would not occur as often with safe ends on both ends of the flue, and not at all with a steel tube. Our practice is to weld on both ends of the flue alternately. A flue put in service to-day, with a safe end at the firebox end, may run from one to two years, and when taken out the bead is cut off the firebox end and the front end cut off with a flue cutter, cutting off about 1½ inches from the end of the flue. Now, before welding this flue we again cut to length for welding and apply the safe-end piece on the opposite end for the firebox, then when the flue is welded we cut to length to go in the boiler, but only have to cut 1 inch off of the original safe end. This flue again goes in service for another year or two, and when flue is again taken out the bead is cut off the firebox end and at the front end with a flue cutter, and is then cleaned and cut for welding. This time we cut the original weld off, and the new piece or safe end is again welded on the front end, which leaves the flue with but two welds, and this operation continues through the life of the flue, and it always gives us a good new safe end for the firebox and the old safe end for the front flue sheet. By following this method we do not know what it is to split the flue at the front end or to damage a flue in the rattler. We have a better flue at all times than a flue showing five or six or more welds. I also believe that if all flues, steel or iron, were taken care of in this manner flues would be less liable to stop up. A great many might object to the cutting of this flue for welding, but unless you do this you cannot tell where the old weld will come at the front flue sheet. And the cost for cutting flues to length for welding is less than one-half cent per flue, or about 50 cents per hundred. I believe if this practice were carried out you would have a flue that was worth that much more to you at all times.

According to reports the steel industry is now operating at about 73 percent of its capacity. The improvement in this business has been manifest during the last month, and steel men seem confident that the improvement is permanent. Orders for railway work are now far below normal, but large equipment orders are pending.